

Unit 7

ETHICS AND OWNERSHIP

27 items · 125 marks

Syllabus sections in this unit

7.1 Ethics and Ownership

Unit 7: Ethics and Ownership

1. M/J 26/P11/Q2/d

- 2 A computer programmer uses language translators. Two types of translator are an interpreter and a compiler.

(d) The completed computer program is distributed under a licence.

Complete the table by writing the name of each software licence that is described.

software licence	description
	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.
	The software is purchased for a fee. There is no access to the source code.
	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.

[3]

2. O/N 25/P11/Q8/b,c

- 8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(b) Give **three** other ethical considerations for a student using their personal computer to connect to the school network.

1

2

3

[3]

(c) Describe **two** social impacts of students using Artificial Intelligence (AI) to complete their homework.

1

2

[4]

3. O/N 25/P12/Q2/a,b

2 (a) Complete the table by describing each term.

Term	Description
Copyright	
Open Source (Initiative)	
Shareware	
Software Licence	

[4]

(b) Describe the purpose of a code of conduct.

[2]

4. O/N 25/P12/Q10/b

10 A programmer is developing a computer program.

(b) The programmer releases the program as Free Software.

Describe what is meant by Free Software.

[2]

5. O/N 25/P13/Q7/a

7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

[2]

6. O/N 25/P13/Q8/c

- 8** Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(c) A programmer is developing an Operating System (OS).

Identify **two** types of software licence the programmer can use to allow other people to edit and redistribute the OS.

1

2

[2]

7. M/J 25/P11/Q4/a

- 4** A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

(a) The new system uses digital cameras and Artificial Intelligence (AI) to identify the customers.

Explain how the new system uses AI to identify each customer.

[4]

8. M/J 25/P12/Q3/a,b,c

- 3** A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

- (a)** Explain how AI is used in the computer program described.

[4]

- (b)** State **two** social benefits of the use of AI in the computer program described.

1

2

[2]

- (c)** The computer program is released under a commercial software licence.

- (i)** Describe the features of a commercial software licence.

[3]

- (ii)** Explain the reasons why an open source software licence might **not** be appropriate for the computer program described.

[3]

9. M/J 25/P13/Q3/d

- 3** A programmer is writing a program in a high-level language.

- (d)** Describe the possible consequences of the programmer **not** joining a professional ethical body.

[3]

10. O/N 24/P11/Q6

- 6** A car park system uses a camera to record the registration number of each car as it enters and leaves the car park.

Explain how artificial intelligence is used in the car park system to identify the car's registration number.

[4]

11. O/N 24/P11/Q7/a,b

7 Software is distributed with a licence.

(a) Give **two** benefits of distributing software using a shareware software licence.

1

2

[2]

(b) Give **two** benefits of distributing software using a commercial software licence.

1

2

[2]

12. O/N 24/P12/Q5/a,b

5 A computer programmer is learning about the importance of acting ethically at work.

(a) Explain the reasons why the computer programmer needs to act ethically towards colleagues and the public.

Colleagues

The public

[4]

(b) The programmer writes a program to be used in a business. The source code of the program can be modified by the business. The business pays the programmer for the maintenance and security updates.

(i) Identify **one** suitable type of software licence for the program **and** describe the reasons for your choice.

Type of software licence

Reasons for choice

[3]

(ii) Explain the reasons why the programmer should copyright their program.

[2]

13. O/N 24/P13/Q5/a,b

5 A computer programmer writes programs that are distributed with a software licence.

(a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	
Users can try the software before buying it. Users may not be able to access all features during the trial.	
Software is usually free of charge. Users can access the source code and alter the program to their needs.	

[3]

(b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

[3]

14. M/J 24/P11/Q5/c(ii)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(c) The bank also needs to keep its customers' data private and secure.

(ii) Customers need to use biometric authentication to access their accounts.
One biometric authentication method is facial recognition.

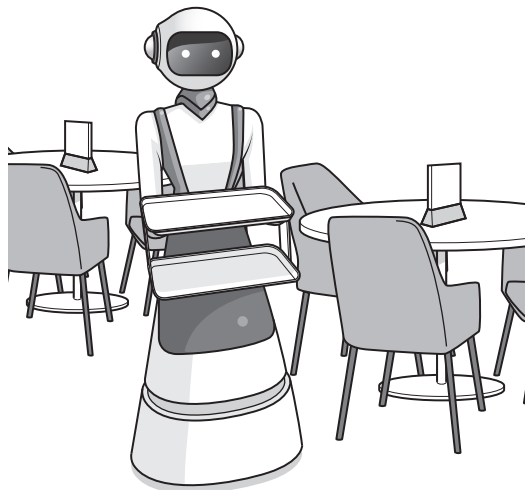
Facial recognition uses Artificial Intelligence (AI).

Describe how AI is used in facial recognition.

[4]

15. M/J 24/P13/Q7/b

- 7 Robots are used to serve food and drink to customers at a restaurant.



- (b) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

[3]

**16. O/N 23/P13/Q6/a,b**

- 6 (a) State **two** benefits to a programmer of distributing a program using a shareware licence.

1

2

[2]

- (b) Explain why it is important for a programmer to join a professional ethical body.

[4]

17. M/J 23/P11/Q4/c

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (c) The CCTV system uses Artificial Intelligence (AI) to identify the presence of a person in the house and to track their movements.

Describe how AI is used in this system.

[3]

18. M/J 23/P12/Q7/d

7 A software developer is working in a team writing a program for a client.

(d) One section of the program being developed will convert user's speech into commands.

Explain how Artificial Intelligence (AI) can be used in this program.

[3]

19. M/J 23/P13/Q3/b

3 A mobile telephone is used to record a video.

(b) The mobile telephone uses a built-in digital camera to record the video.

The digital camera automatically focuses on the faces of people.

Explain how Artificial Intelligence (AI) is used by the camera to automatically focus on the faces of people.

[3]

20. M/J 23/P13/Q5/b

5 A programmer is developing a computer game in a high-level language to sell to the public.

(b) The programmer needs to publish the game under a software licence so that it can be sold to the public.

Identify the **most appropriate** type of software licence for the game **and** justify your choice.

Licence

Justification

[4]

21. O/N 22/P12/Q9

9 One use of Artificial Intelligence (AI) is for facial recognition software.

Describe the social impact of using facial recognition software to identify individuals in an airport.

[2]

22. O/N 22/P13/Q8/a,b

- 8 (a) (i) Explain why some programs are distributed under an open source licence.

[2]

- (ii) Explain how a programmer benefits from distributing a program under a commercial licence.

[2]

- (b) A commercial program for a vehicle repair garage includes an Artificial Intelligence (AI) module that can diagnose faults and suggest repairs.

Describe **one** economic impact the AI module may have on the garage.

[2]

23. O/N 21/P11/Q4/a,b(i),c

- 4 Francis is starting his first job as a software developer for a multinational company.

- (a) Francis has been advised to join a professional ethical body.

Describe the benefits to Francis of joining a professional ethical body.

[3]

- (b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

- (i) Describe the ways in which Francis can act ethically in this situation.

[2]

- (c) Francis is part of a team writing a program. He finds an error in part of the program that has already been tested. He decides not to tell anyone because he is worried about the consequences.

Explain the reasons why Francis acted unethically in this situation.

[2]

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Explain the reasons why Francis acted unethically in this situation.

[2]

25. M/J 21/P11/Q1/d

1 Anya scans an image into her computer for a school project.

(d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

26. M/J 21/P12/Q2/a,b,c

2 Aisha manages a team of software developers.

- (a) Explain the reasons why it is important that Aisha acts ethically in relation to her team members.

[2]

- (b) The team are developing a computer game where the user plays a board game (such as chess) against the computer.

Describe how the computer would use Artificial Intelligence (AI) to play the board game.

[3]

- (c) The final game will be released under a licence.

Tick (✓) **one or more** boxes in each row to identify the licence(s) each statement describes.

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code				
The user must always pay before being able to use the software				
The user can redistribute the software				
The user always gets a trial period				

[4]

27. M/J 21/P13/Q1/d

1 Anya scans an image into her computer for a school project.

- (d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

Answers Section

1. M/J 26/P11/Q2/d

2(d)	1 mark for each correct licence		3	Do not accept copyleft for open source Do not accept freeware for open source
	Software licence	Description		
	Shareware	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.		
	Commercial // proprietary	The software is purchased for a fee. There is no access to the source code.		
	Open source initiative // free software	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.		

2. O/N 25/P11/Q8/b,c

8(b)	1 mark per bullet point, max 3 marks e. g. - Students should consider privacy issues - Students should not make inappropriate use of social networking // Students should not engage in cyberbullying - Students should not copy work that is not their own - Students should abide by the school rules/guidelines for accessing the network - Students should only use the school network for legitimate purposes - Students should not hack / crack other computers	3
8(c)	1 mark per bullet point, max 4 marks e. g. Negative Impacts - If students use AI to shortcut learning, they may miss out on developing reasoning and problem-solving skills // Students may not be learning anything // It could lead to a decline in specific skills - It could have negative impacts on student communication // heavy reliance on AI may limit opportunities for collaboration, teamwork, and face-to-face communication - Emphasises the digital divide// some students may not have access to the technology - The AI answer might not be correct leading to mis-information - Increased screen time and isolation could contribute to anxiety or loneliness Positive Impacts - Students given additional support may do better - Pupils who struggle in traditional settings may gain self-esteem through AI-assisted learning at their own pace Long-Term Impacts - Protection of intellectual property // ethical issues and considerations - Schools may need to redesign curricula and/or assessment to integrate AI	4

3. O/N 25/P12/Q2/a,b

2(a)	1 mark for each correct row, max 4 marks	4										
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2(b)	1 mark per bullet point, max 2 marks e.g. • to create a safe, respectable and professional working environment for all employees • to ensure that employees understand the expectations • ... and the consequences of their actions • to provide a clear outline of what is considered good practice • so there are clear rules to follow in certain situations • to protect the reputation of the organisation // maintain public trust	2										

4. O/N 25/P12/Q10/b

10(b)	1 mark per bullet point, max 2 • The software is free of restrictions • The software is not necessarily free of charge • The <u>source code</u> comes with the program // A user can edit the <u>source code</u> to add functionality / suit their needs • Users can share the software with others	2
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5. O/N 25/P13/Q7/a

7(a)	1 mark per point, max 2 marks e.g. • The learning experience could be improved • ... by identifying students who are struggling • and employing more personalised learning • There could be early Intervention • ... to determine which students require extra support • Teachers and students could understand the challenges better • ...and identify optimal times for learning • There could be privacy concerns • students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • due to constant monitoring and pressure • contributing to student stress and anxiety	2
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6. O/N 25/P13/Q8/c

8(c)	1 mark per bullet point, max 2 marks • Open Source (Initiative) • Free Software (Foundation)	2
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7. M/J 25/P11/Q4/a

4(a)	1 mark each to max 4 e.g. • Uses image recognition / facial recognition • Measures the distance between facial features of the customer • and stores the values in a database of user information • An image is captured using the digital camera • the AI identifies that the image is a face • by analysing the pixels to find patterns • the distance between features is calculated and compared to those in database	4
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8. M/J 25/P12/Q3/a,b,c

3(a)	1 mark each to max 4 e.g. • Uses image recognition // optical character recognition • to analyse a photograph/pixels to identify the location of characters within an image • in order to convert the pattern of pixels into individual characters • Uses natural language • to combine the characters into words • and compare the words to database to identify a match in the language of the writing • it may also predict what is missing / incomplete / from what has been read • then compare the words to a database of the user's chosen language to find a match and translate • Uses text-to-speech • to generate the audio waveforms corresponding to the words	4
3(b)	1 mark each to max 2 e.g. • It can help people with visual impairment to identify which products to purchase • It can help people who cannot understand the language • It can help to overcome learning / reading difficulties	2
3(c)(i)	1 mark each to max 3 • The program is usually sold for a fee • Users have no access to source code • It protects the intellectual rights of the programmer / developer • Users cannot legally alter or share the program • There are limitations on the use of the software e.g. number of times it can be installed • Support and maintenance is usually provided e.g. updates, bug fixes etc.	3
3(c)(ii)	1 mark each to max 3 e.g. • The source code would be accessible therefore it could be changed • which might lead to the program outputting an incorrect response • so people could purchase the wrong item • which might have serious consequences // by example e.g. allergies	3

9. M/J 25/P13/Q3/d

3(d)	1 mark each to max 3 e.g. • There is no / limited access to legal advice • in case action is taken against them • There are fewer networking opportunities • so they could miss out on contacts / jobs • There would be less access to training • There would be no clear laid out ethical guidelines • and / or people to discuss potential ethical problems with • possibly leading to inappropriate / unethical actions • which might lead to legal proceedings / recourse	3
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10. O/N 24/P11/Q6

6	1 mark for each bullet point (max 4) e.g. • It uses image recognition • The pixels of each image from the camera are stored • ... and matched to the expected shape/size/colour of registration number • It uses optical character recognition • The pixels within the registration number area identified are analysed • ... and compared to expected characters/letters/numbers	4
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11. O/N 24/P11/Q7/a,b

7(a)	1 mark for each bullet point (max 2) • The user can test/trial the program before purchasing to see whether it meets needs • There is no initial cost • Users who try the software may spread the word so the developer gets more custom • The developer can get feedback to improve the program	2
7(b)	1 mark for each bullet point (max 2) • The owner/developer can earn a fee • Comprehensive support is usually offered so problems can be easily solved • The software is usually well tested so few bugs	2

12. O/N 24/P12/Q5/a,b

5(a)	1 mark for each bullet point max 2 for public max 2 for colleagues e.g. Colleagues: The programmer should: • Treat colleagues fairly and avoid acts of discrimination • Be prepared to accept / offer critique • ... so that technical work can be improved • Credit the contributions of colleagues • ... so that they feel valued / respected • Help / train colleagues etc. e.g. The public: The programmer should: • Maintain the health, safety and welfare of the public • ... so as not to endanger anyone • Be honest / realistic in making claims about the software • ... so that public trust is maintained • Maintain the security of the public's data etc.	4
5(b)(i)	Type of software licence 1 mark for • Open Source Initiative // Free Software Foundation Reason for choice: 1 mark for each bullet point (max 2) • Allows program source code to be examined • ... so that the business can adapt/modify the program to meet their needs • Allows the programmer to earn money for the software and updates	3
5(b)(ii)	1 mark for each bullet point (max 2) • To identify themselves as the owner/author // To gain formal recognition of ownership • To allow for legal consequences if anyone steals/copies it • To restrict competition	2

13. O/N 24/P13/Q5/a,b

5(a)	1 mark for each correct licence	3								
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5(b)	1 mark for each bullet point (max 3) e.g. • The professional ethical body has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so the computer programmer does not have to decide what is ethical it's written down • Clients/staff know the computer programmer is reputable • ... there may be a test/requirement for entry • ... giving recognition of the computer programmer's skills/knowledge • The professional ethical body provides help and support • ... for example, if the computer programmer needs legal advice • The professional ethical body runs training courses • ... to keep the computer programmer's skills up to date	3								

14. M/J 24/P11/Q5/c(ii)

5(c)(ii)	1 mark each to max 4 : e.g. • Captures an image of the face • Uses image recognition • Trained to identify the features of a face in an image • ... using a large number of images • Analyse images for facial features • Uses the probability of a match	4
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15. M/J 24/P13/Q7/b

7(b)	1 mark each to max 3 : e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • ... programmed action(s) for matching sentence(s) are performed	3
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16. O/N 23/P13/Q6/a,b

6(a)	1 mark for each bullet point (max 2). • Users are able to trial the program and may offer the programmer constructive feedback • More people might try the program because the trial is free of charge • Allows bugs to be found and corrected on a wide range of computer system configurations improving the programmer's original program • Users that find the trial useful will buy the program so programmer gets income • Allows the program to be copyrighted and so protects the programmer's intellectual property rights // no illegal modification allowed	2
6(b)	1 mark for each bullet point (max 4). • The programmer has ethical guidelines to follow • ...so purchasers of programs know the standards being applied • ...so programmer does not have to decide what is ethical or not • Clients know programmer is reputable • ...recognition of programmer's skills/knowledge • There may be an entry requirement / exam • ... so clients know that the programmer is competent • Professional ethical body provides help and support • Such as legal advice // by appropriate example • Enables programmer to attend the ethical body's training courses • ...to keep skills up to date // to develop skills	4

17. M/J 23/P11/Q4/c

4(c)	1 mark each to max 3 Examples: • Uses image recognition • Monitors every image taken to identify matching images/shapes/features to a 'person' ... • ... starts recording to secondary storage/permanently when a person is identified • System identifies direction of movement of person and uses this to decide where/how to move the camera/record • System identifies other cameras to start recording based on direction of movement	3
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18. M/J 23/P12/Q7/d

7(d)	1 mark each to max 3 • Uses speech recognition • ... which identifies key phrases / words spoken • ... and matches these to a database • ... and generates the most likely sentence / command / word	3
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19. M/J 23/P13/Q3/b

3(b)	1 mark each to max 3 Examples: • Scans the scene in real time • Identifies if there are faces in the image • Uses facial recognition • ... uses image recognition • ... takes each frame individually • ... analyses the pixels • ... stores pattern for a face • ... looks for patterns that match/come close to the pattern for a face • Camera focuses on the pattern identified	3
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20. M/J 23/P13/Q5/b

5(b)	1 mark for appropriate licence; 1 mark for each point to max 3 - Commercial software licence - User has to pay for the product so the programmer can gain an income - Enables the program to be copyrighted ... so the user cannot legally edit the program // the programmer retains control over product ... and can take legal action against people who attempt to illegally copy it /sell it on - Shareware licence - Enables the program to be copyrighted - The user cannot legally edit the program so the developer retains control over product - User can try the program for free and then pay for the full game which allows the programmer to gain an income - so more people can experience it and therefore be more likely to buy it	4
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21. O/N 22/P12/Q9

9	1 mark for each bullet point (max 2). For example: - incorrect recognition of faces leads to mistakes such as ... access to facilities / systems may be denied - privacy issues / people do not like data being stored - individuals will feel safer ... there might be a reduction in crime - faster boarding - catching criminals	2
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22. O/N 22/P13/Q8/a,b

8(a)(i)	1 mark for each bullet point (max 2): - to allow users to customise the code - to allow errors to be reported / identified / fixed by users - to allow additional features to be added to the code - to allow for collaboration	2
8(a)(ii)	1 mark for each correct point (max 2) Example: - enables the program to be copyrighted - prevents illegal changes to the program / protects the source code - prevents illegal copies of the program being made - a fee can be charged for the program	2

8(b)	1 mark for a correct economic impact and 1 mark for corresponding description Example: • reduce costs to the garage • ... because less time taken for diagnosis • increase profits for the garage • ... as technicians spend more time repairing, so completing more jobs in a day • decrease costs passed to customer • ... so garage may gain customers • profit margins can be reduced • ... because program may be expensive to buy / maintain / update	2
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23. O/N 21/P11/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

4(c)	1 mark per bullet point to max 2 e.g. • He didn't act in best interest of product • ... because the product might fail because he didn't report the error • He didn't act in best interest of client • ... because if the product does not work then they have been let down because he didn't report the error • He didn't act in the best interest of the profession • ... he is letting his profession down because he didn't report the error • He didn't act in best interest of the company • ... not correcting the error early could lead to later problems	2
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24. O/N 21/P13/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

4(c)	1 mark per bullet point to max 2 e.g. • He didn't act in best interest of product • ... because the product might fail because he didn't report the error • He didn't act in best interest of client • ... because if the product does not work then they have been let down because he didn't report the error • He didn't act in the best interest of the profession • ... he is letting his profession down because he didn't report the error • He didn't act in best interest of the company • ... not correcting the error early could lead to later problems	2
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25. M/J 21/P11/Q1/d

1(d)	1 mark per bullet point to max 2 • The formal and legal rights to ownership // intellectual property rights • Protects against unauthorised reproduction of work • Provides for legal right of redress	2
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26. M/J 21/P12/Q2/a,b,c

2(a)	1 mark per bullet point to max 2 • To make sure the team members feel valued • To get the best work out of the team • To enable them to work well together • To enable them to create the best product for the client	2
2(b)	1 mark per bullet point to max 3 • The rules / past moves / decision making algorithms of the game will be stored • The AI program is trained, by playing many times • AI will look (ahead) at possible moves • ... and/or analyse the pattern of past choices • ... and choose the move most likely to be successful • Computer could learn how to improve // learn from previous mistakes • ... by storing the positive/negative result of choices • ... and changing its future choices	3

2(c)

1 mark for each correct column

4

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code	✓	✓		
The user must always pay before being able to use the software				✓
The user can redistribute the software	✓	✓	✓	
The user always gets a trial period			✓	

27. M/J 21/P13/Q1/d

1(d)	1 mark per bullet point to max 2 • The formal and legal rights to ownership // intellectual property rights • Protects against unauthorised reproduction of work • Provides for legal right of redress	2
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